A MORPHISM OF FINITE PRESENTATION IS OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement consists of two sentences without an enumeration; both are inside the counted statement span and the proof handles them in turn.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The source-and-target presentation of local finite presentation. Lemma lemma-local-source-target, used at proof lines 5751-5754. Produces the scheme morphism h and the surjective etale a.
- The scheme case: locally of finite presentation implies locally of finite type. Morphisms, Lemma morphisms-lemma-finite-presentation-finite-type, used at proof lines 5754-5756. Gives local finite typeness of h.
- Local finite typeness is read off from a source-and-target presentation. used at proof lines 5756-5757. Concludes the local statement for f.
- Finite presentation includes quasi-compactness. used at proof lines 5758-5759. Upgrades the local statement to the global one.

Source reference. Original lemma 03MJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4751–4751. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01TW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2389–2389. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *A morphism which is locally of finite presentation is locally of finite type. A morphism of finite presentation is of finite type.*

Proof. Let $f : X \to Y$ be a morphism of algebraic spaces which is locally of finite presentation. This means there exists a diagram as in Lemma 03MJ with h locally of finite presentation and surjective vertical arrow a. By Morphisms, Lemma 01TW h is locally of finite type. Hence $X \to Y$ is locally of finite type by definition. If f is of finite presentation then it is quasi-compact and it follows that f is of finite type. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.